HARDIK KUMAR SINGH

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EDUCATION

Motilal Nehru National Institute of Technology, Allahabad **CPI: 8.27**
B.Tech in Computer Science and Engineering *Dec 2021 – May 2025*

EXPERIENCE

Grid Dynamics Bengaluru, India
Software Engineer *Jan 2025 – Present*

VISA VASPD - Acceptance Solutions

- Developed **Java** backend enhancements for merchant onboarding and payment workflows, resolving synchronization issues through cross-service integration and database validation.
- Investigated production issues using **Splunk, Grafana, and Oracle SQL Developer**, performing root cause analysis across application, API, and database layers.
- Improved payment API usability by replacing generic validation errors with **domain-specific error enums** and scenario-based messages for **tokenization workflows**, adopted by the Product team.
- Identified terminology inconsistencies in Virtual Terminal **pre-authorization workflows** and proposed scheme-aligned transaction naming adopted by the Product team to improve product clarity.

AI-Powered Talent Acquisition Platform

- Developed backend systems for an **AI-driven Talent Acquisition Platform**, reducing manual candidate screening effort by **90%**.
- Built asynchronous REST APIs using **FastAPI** and integrated **Google Vertex AI** with **OpenAI GPT-4** for semantic CV-to-JD matching, improving recruiter efficiency by **35%** and matching accuracy by **35%**.
- Designed an autonomous candidate scoring engine achieving a **91% reduction in screening time** while improving shortlisting consistency by **30%**.
- Containerized backend services using **Docker** and deployed on **Google Cloud Run** with **Cloud SQL (PostgreSQL)**, reducing operational overhead by **25%**.

LLM-Powered Learning Assistant Demo

- Led backend development of an **LLM-based Learning Assistant**, transforming **90%** of diverse learning content into interactive learning experiences.
- Built **35+ REST APIs** using **FastAPI**, integrating **PostgreSQL**, **Redis**, and **ChromaDB**, while implementing secure code execution using **Firecracker microVMs**, reducing execution-level security risks by **80%**.
- Integrated AI chatbot, Q&A generation, and automated code evaluation with cross-functional teams, increasing learner engagement by **30%**, while containerizing services using **Docker** for consistent deployments.

PROJECTS

OptiVision | *Python, TensorFlow, OpenCV, Jetson Nano* *Jan 2024 – May 2024*

- Developed a real-time multi-camera object detection pipeline using **CenterNet ResNet-101**, achieving **32.4 mAP** with **34 ms** inference latency on Jetson Nano.
- Processed **5+ concurrent RTSP streams** for multi-object detection and tracking, achieving **85%+ detection accuracy** while maintaining real-time performance.
- Established an event logging pipeline capturing **10k+ object detections** to support downstream analytics and debugging.
- Implemented a **TF-IDF**-based search assistant for querying historical detection events without manual log inspection.

TECHNICAL SKILLS

Programming Languages: Java, C++, SQL
Backend Development: Spring Boot, FastAPI, REST APIs, Microservices, Distributed Systems
Databases: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB
Messaging & Streaming: Apache Kafka
Cloud & DevOps: Google Cloud Platform (Cloud Run, Cloud SQL), Amazon Web Services (EC2), Docker, Kubernetes
AI-Assisted Development: Cursor, Claude Code, OpenAI Codex
Developer Tools: Git, GitHub, Maven, JUnit, Mockito, Postman, Splunk, SQL Developer, IntelliJ IDEA, Visual Studio Code
Methodologies: Agile, Scrum

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)**